

# Determination of the Coherence Length of a Light Emitting Diode

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## ABSTRACT

*The coherence length of a red LED was determined to be  $(6.2 \pm 0.3) \mu\text{m}$  via spectroscopic analysis, with a wavelength of  $(678 \pm 14) \text{ nm}$ . This method was used after the intended procedure failed to produce results due to limitations in the precision with which the interferometer could be calibrated. An analytical solution for the coherence function and envelope of any Gaussian polychromatic light source was also derived.*

## 1 Introduction

THE phenomenon of wave interference is a ubiquitous feature of undergraduate physics courses worldwide. In the context of physical optics, the discussion is almost always limited to the self-interaction of monochromatic light alone. Despite polychromatic light sources being far more common in the Universe, the consideration they are afforded is not often in proportion to their prevalence.

Of course, including multiple wavelengths in the analysis of light brings with it extra complications. In particular, waves of different wavelengths typically occupy different phases when

considered even at the same point in space, and will thus interfere to different extents there. It is this effect which complicates this interference beyond that of monochromatic light, where only the contribution of a single wave need be examined.

In the near-field, similar constituent wavelengths of polychromatic light will be of similar phase at every point in space, as the difference in phase caused by their different wavelengths over a given distance won't be substantial. If this light is allowed to travel into the far-field, however, then even very similar wavelengths will drift to completely different phases at the same point — the different frequencies which compose the light appear to become incoherent.

A distance, called the *coherence length*, at which the waves of different frequencies transition from marching in lock-step to existing independently, can be defined and will encode crucial information about the light being analysed, including the bandwidth of the source (see subsection 2.6). The purpose of the experiment in this study is to determine this coherence length in the case of a light emitting diode by means of a Michelson interferometer.

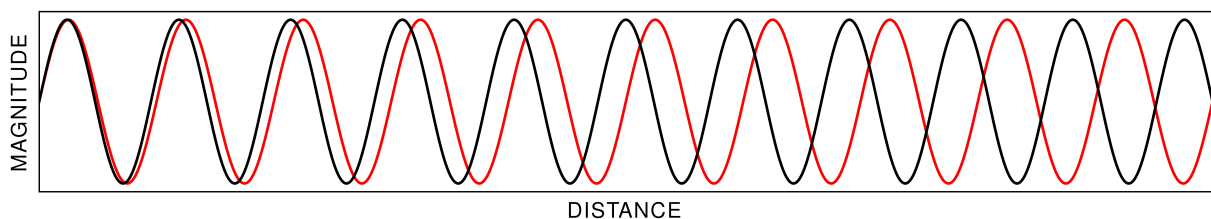


Figure 1: Waves of similar frequency are coherent in the near-field but not in the far-field.

## 2 Theory

### 2.1 A Note on Convention

The definition of coherence length changes depending on which field is being considered. In some cases, the coherence length is the distance which two waves must travel such that the intensity of their interference fringes decreases by a factor of  $\frac{1}{e}$  (Conte et al., 2018). In many optical studies, however, the coherence length is defined as the distance by which a mirror in the Michelson Interferometer (subsection 3.1) must be moved to reduce the interference intensity by a factor of  $\frac{1}{2}$  (Dubey et al., 2016).

This second definition proves to be more natural for the experiment outlined in this study. Also note that the difference in mirror displacement (DMD) is the difference in length from the beam splitter to each mirror — half of the optical path difference (OPD). This metric for the coherence length is used widely in similar studies, and any results quoted from literature will be in this convention as well.

### 2.2 Monochromatic Interference

THE manor in which monochromatic light interferes with itself is of upmost relevance when considering the interference of polychromatic light, and so it's worth some discussion here. To investigate the interference of monochromatic light, the Michelson interferometer is the standard apparatus, and is the device which will be at the crux of the experiment conducted in this study also.

The interferometer splits an incoming beam of light into two beams which each travel along perpendicular paths of length  $L_1$  and  $L_2$ . The two beams then recombine at the same point they were split, and the resulting beam depends on the phase difference between the returning beams, which is itself dependent on the OPD  $L_2 - L_1$ . If one beam travels exactly an integer multiple of wavelengths further than the other, then a construction of magnitude will be observed upon their recombination as the waves are periodic. If

this path length difference is a half-integer multiple of the wavelength instead, then the two beams will entirely destroy each other as the peaks of one wave combine with the troughs of the other.

### 2.3 Problems with Interference

IN the common explanation of monochromatic interference given above, the principle of operation of the Michelson interferometer seems deceptively simple. In reality, there are a few complications which must be considered before the theory even permits a discussion of polychromatic light.

Firstly, the polarisation of the light is a concern. If the light isn't polarised when it's emitted from the source, then the simple model of crests and troughs lining up doesn't work, as the chances of two different waves being polarised by chance is minimal. So, there must be some way to ensure that any two waves of light which interact with each other at the beam splitter will be polarised in the same way.

Secondly, light does not travel as a plane wave, instead travelling classically in wave packets. If these wave packets are emitted from the source spontaneously, then the probability of two wave packets arriving back at the recombination point simultaneously is too small to predict a sustained interference pattern. For this to be resolved, there must be an assurance that for every wave which arrives back at the beam splitter, there will be another arriving along the other path at the same time.

These problems are often overlooked in the case of monochromatic interference, as the light source used is often a laser which will produce a high intensity and highly polarised stream of photons anyway.

### 2.4 The Beam Splitter

A closer investigation of the operating principles of the beam splitter will solve the problems previously mentioned. To elucidate this, consider what happens if a single photon is sent into the beam splitter. As the photon is a fundamental particle in the standard model, it can't be

bifurcated — and yet single-photon interference is an experimentally established phenomenon.

The photon can interfere with itself because, while the particle cannot be split, the particle's wave function *can* be diffracted. The wave function then travels along both arms of the interferometer, reflects back and recombines again at the site of diffraction. Thus it is irrelevant whether the photon is part of a larger beam or not, each photon can only meaningfully interfere with itself. This means that the two interfering waves will have the same polarisation because they both result from the same diffraction event, the same is true for the issue of temporal coincidence of waves at recombination.

This now opens up the possibility of polychromatic interference, as a source of multiple wavelengths can be modelled as many monochromatic sources contributing simultaneously. As each wave packet will only ever interact with itself, there is no difference at the photonic level.

## 2.5 Polychromatic Interference

THE Michelson interferometer allows polychromatic light to interfere linearly in the mathematical sense. The interferential effect of many monochromatic light sources is the same as the sum of the effects of each source individually and, as stated before, a polychromatic source can be modelled as the sum of many monochromatic sources. This additive property means that the interference of *any* light entering the interferometer can be predicted analytically if the spectral power distribution of the incident light is known. Beyond that, a knowledge of the interference of monochromatic light is all that is needed.

When the OPD, and therefore DMD  $\delta$ , is zero, then all wavelengths of light will constructively interfere, as  $\delta = 0$  is technically an integral multiple of any wavelength  $\lambda$ . Take some wavelength  $\lambda_0$  in the spectral power distribution of the LED, photons of this wavelength will constructively interfere when  $\delta = n\lambda_0 : n \in \mathbb{Z}$ . If a photon has a slightly smaller wavelength, then it will constructively interfere at  $\delta < \lambda_0$ , and

the opposite is true of those photons with a slightly longer wavelength. As the path length difference is made larger, the photons of higher frequency are ahead of phase relative to photons of wavelength  $\lambda_0$  undergoing total constructive interference, and the opposite is true for photons of smaller frequency. If  $\delta$  is large enough, then photons of similar wavelengths will no longer interfere in the same way around the same point, they will become incoherent with each other.

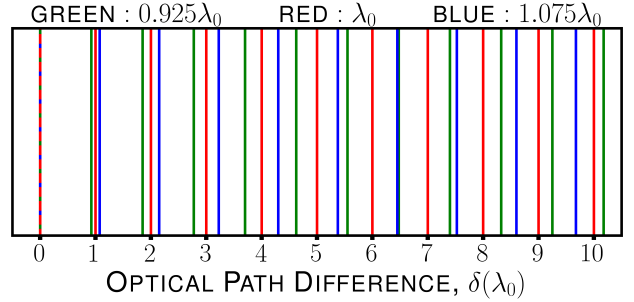


Figure 2: The OPDs at which photons of three similar wavelengths undergo total constructive interference become less similar as the magnitude of those OPDs increases.

In Figure 2, this effect can be seen. While the various wavelengths of polychromatic light will interfere as a "group" at small OPDs, there comes a point at which an interference fringe of some order of one wavelength overlaps a lower order fringe of a different wavelength ( $\delta \approx 6.5\lambda_0$  in Figure 2). Beyond this point, the polychromatic light stops interfering like monochromatic light, and the interference of many wavelengths averages out to give one constant luminosity. Furthermore, the distance to this point will evidently become smaller as the difference between the smallest and largest wavelength components of the light (*i.e.* the bandwidth) increases, as the lines in Figure 2 will spread out quicker as this difference becomes larger.

Almost all detectors, including the one used in this study, measure the time-averaged intensity of the light incident upon them. The intensity is simply the square of the magnitude of the electric field at the detector. If the post-interference polychromatic light is allowed to strike the detector, then the resulting intensity can be found by summing the contribution from each wavelength and

squaring that sum. Then, the average of this intensity can be found over a long time and that will be what the detector measures. The actual output of the detector will be a voltage which is proportional to this measurement (Enlitech, 2024).

$$V \propto \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T ||\mathbf{E}||^2 dt \quad (1)$$

Figure 3 shows this measurement as a function of the difference in mirror displacement of the two paths in the interferometer. The sinusoidal nature of this "coherence function" is similar to that of monochromatic light, but the amplitude decays as the mirror displacement is increased due to the decoherence previously explained. The envelope inside of which the function sits, called the "coherence envelope" is determined by the coherence length of the light source alone. The goal of this experiment is to find this interference profile for an LED and thereafter determine its coherence length.

## 2.6 Coherence Time and Bandwidth

NOT only do the coherence functions in Figure 3 contain information about the coherence length of the light source, but it also contains information about the spectral power distribution of the light source. In fact, the coherence envelope is the connection between the bandwidth and the coherence length.

The relation between the coherence length and bandwidth can be derived by introducing another related quantity: the "coherence time"  $\tau_c$ . The coherence time can be defined as the amount of time it takes for two waves with frequencies at opposite ends of the bandwidth to overlap each other. Mathematically, let one wave have a frequency  $\nu_1$  and the other have a frequency  $\nu_2 = \nu_1 + \Delta\nu$ , the coherence time is the time it takes for the second wave to achieve a phase of  $\phi_2 = \phi_1 + 2\pi$ , where  $\phi_1$  is the phase of the first wave. Let the initial conditions be such that the phase  $\phi$  of a wave is given by  $\phi = 2\pi\nu t$ . The phase dif-

ference  $\Delta\phi$  after a coherence time  $\tau_c$  will be  $2\pi$ .

$$\Delta\phi = \phi_2 - \phi_1 \quad (2)$$

$$= 2\pi\nu_2 t - 2\pi\nu_1 t \quad (3)$$

$$= 2\pi\Delta\nu t \quad (4)$$

$$2\pi = 2\pi\Delta\nu\tau_c \quad (5)$$

$$1 = \Delta\nu\tau_c \quad (6)$$

So,  $\tau_c$  is related to  $\Delta\nu$  by Equation (7).

$$\tau_c = \frac{1}{\Delta\nu} \quad (7)$$

The coherence length  $L_c$  is the distance over which a phase difference of  $2\pi$  emerges, and so can be related to the coherence time  $\tau_c$  by Equation (8).

$$\frac{L_c}{c} = \tau_c \quad (8)$$

Then, the relation between the coherence length and bandwidth may be revealed by combining Equation (7) and Equation (8).

$$L_c = \frac{c}{\Delta\nu} \quad (9)$$

By Equation (9), the coherence length has an inverse proportionality to the bandwidth and the constant of proportionality is the speed of light. This enables the value of coherence length to be cross-checked with a spectrometer and Equation (9).

## 2.7 The Coherence Function $I(\delta)$

THE coherence function describes the intensity at the photodiode for a given difference in mirror displacement  $\delta$ , assuming the spectral power distribution (SPD) is Gaussian. The coherence function and the envelope which enclose it contain vital information about the source whose light is interfering, including its coherence length. The coherence function  $I(\delta)$  and its envelope  $I_{\text{env}}(\delta)$  are derived in Appendix A.



$$I(\delta) = I_0 \left[ 1 + \cos(k_0 \delta) \exp \left( -\frac{\sigma_k^2}{2} \delta^2 \right) \right] \quad (10)$$

$$I_{\text{env}}(\delta) = \exp \left( -\frac{\sigma_k^2}{2} \delta^2 \right) \quad (11)$$

where

$I_0$  : The intensity at the photodiode for  $\delta = 0$ .

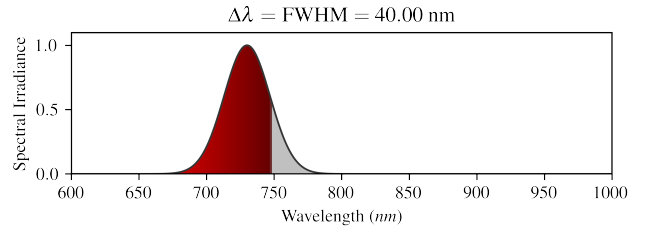
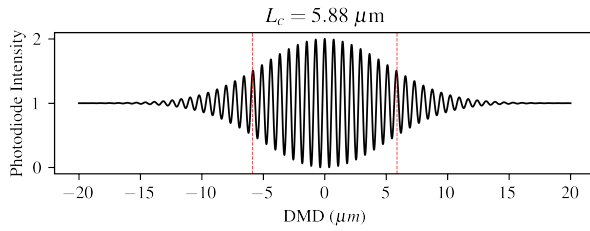
$k_0$  : Centroid wavenumber of the SPD.

$\sigma_k$  : Standard deviation of the SPD.

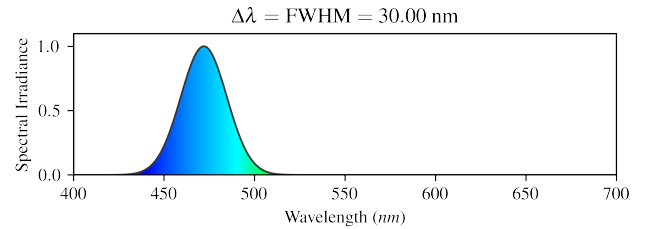
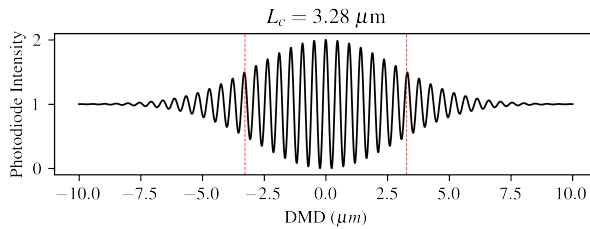
As the coherence function fully describes the interference of polychromatic light on the optical axis, inspecting the properties of the function will give a powerful insight into expected behaviour of the experiment.

Figure 3 shows coherence functions and their respective SPDs, with different  $I_0$ ,  $k_0$  and  $\sigma_k$  parameters. Evidently, those light sources with smaller wavelengths also have interference patterns with smaller wavelengths too. Equation 10 shows that the wavenumber of the interference pattern is the centroid wavenumber  $k_0$  of the SPD. The visualisations in Figure 3 also empirically verify Equation 9, the coherence length of the light source evidently has some inverse relationship to the full-width at half-maximum (FWHM) of the source.

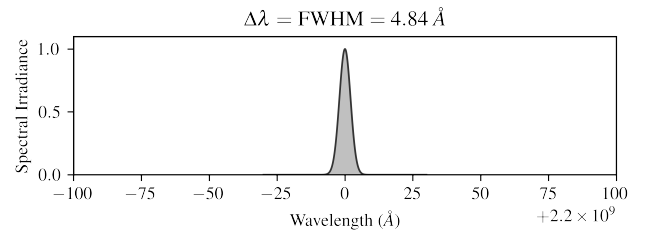
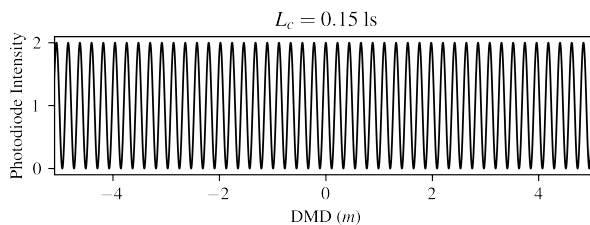
An extreme example of this is shown in Figure 3c. The hydrogen maser uses the hyperfine transition of ground state hydrogen to emit radio waves at  $\lambda = 21$  cm. It emits this wavelength as an extremely narrowband signal, leading to a very long coherence length — on the order of light seconds.



(a) A near-infrared LED. See (TechLed, n.d.) for data.



(b) A carbon quantum dot. See (Yuan et al., 2018) for data.



(c) A hydrogen maser — note the change of units. See (Vremya-CH, n.d.) for data.

Figure 3: The interference patterns and spectral power distributions of named light sources.

### 3 Experimental Procedure

#### 3.1 Overview of Method

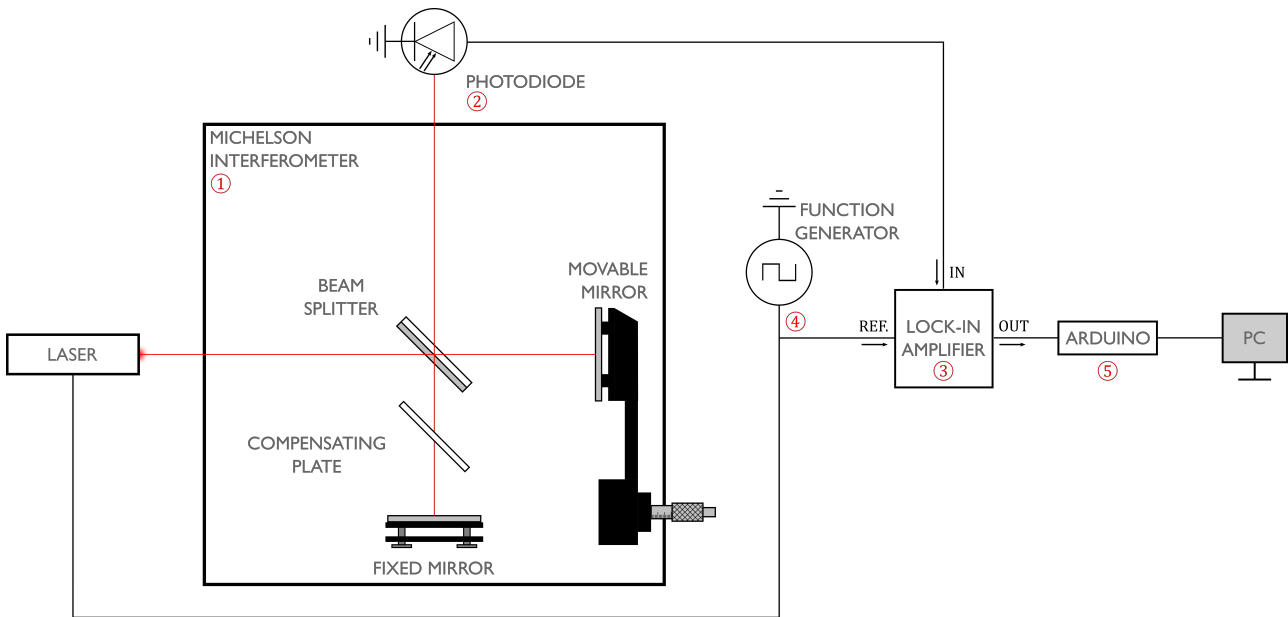


Figure 4: A schematic of the experimental setup with a laser as the light source.

DISPLAYED in Figure 4 is the complete schematic of the experiment. Here, a short summary with numbers corresponding to the above figure will be given to introduce the experiment before going into more detail.

1. The Michelson interferometer (a Henschke 650265 model) split the polychromatic light along two paths, and allowed each frequency to interfere with itself upon recombination. The difference in mirror displacement  $\delta$  was changed and measured via the micrometer.
2. The post-interference light then fell upon the photodiode, which produced a DC voltage proportional to the time-averaged intensity of the light, an expression for which was given in Equation 1. While the Michelson interferometer will produce off-axis rings, only the measurement of the on-axis intensity is relevant.
3. The lock-in amplifier (an LIA-MV(D)-200 series) amplified the small voltage produced across the photodiode by the LED.
4. The function generator provided a square-wave signal at 200 Hz to both the LED and the reference connection of the lock-in amplifier. This chopping of the light signal while simultaneously sending that signal into the lock-in amplifier as a reference allowed only the post-interference light from the LED to be amplified.
5. The Arduino microcontroller read in the voltage data from the lock-in amplifier at a high temporal resolution (100 Hz) and passed it into the PC, where the data was parsed into a Comma Separated Values (csv) file via Microsoft Excel's Data Streamer add-in. Once saved as a csv file, the data could then be plotted using Excel or Matplotlib.

### 3.2 Calibrating the Interferometer

**B**EFORE conducting the experiment, the accurate collimation of each optical component in the Michelson interferometer was crucial. The interferometer was the first device through which error could be introduced, and every subsequent apparatus would suffer if the output of the first was inaccurate. (Henschke, 2005) gives precise instructions on how to collimate the model of precision interferometer used in this experiment. Importantly, after the interferometer was collimated, the screen was lowered and the photodiode placed precisely in the middle of the annular interference pattern, this ensured that the voltage measured was proportional to the *on-axis* time-averaged intensity.

Due to the delicate alignment of the Michelson interferometer and the four-week duration over which the experiment was conducted, the device had to be re-calibrated twice. Fortunately, all measurements were taken within a short time period, and so this re-collimation did not affect the results.

### 3.3 Tuning the Lock-In Amplifier

**W**HILE the lock-in amplifier was a great boon in the context of this experiment, it requires a complex calibration routine to work at full efficiency. On the face of the amplifier are four hexadecimal dials and four DIP switches, enabling  $2^{20}$  (just over one-million) tuning permutations. This large multiplicity of options means that it was not realistic to test every permutation. Instead, the settings recommended by (Femto, 2018) were used to find a state close to the desired tuning, then the area around that permutation was probed to find the ideal settings for the amplifier. The settings which were ultimately used during the experiment are detailed in Table 1.

Table 1: Tuning settings of the lock-in amplifier

| HEXADECIMAL   |   | DIP            |    |
|---------------|---|----------------|----|
| Time Constant | 5 | Dyn. Reserve   | H  |
| Sensitivity   | B | Mode           | 1f |
| Phase Rough   | 0 | PLL            | S  |
| Phase Fine    | 3 | Ref. Threshold | 2V |

During the process of finding the tuning values in Table 1, an oscilloscope was used to measure both the input and output voltages of the amplifier simultaneously. This allowed the signals to be compared in real-time as the settings were adjusted, facilitating the maximisation of the gain and minimisation of erroneous artefacts in the output.

The output itself was taken from the MAGNITUDE terminal of the amplifier, as the other outputs, QUADRATURE and IN-PHASE only output orthogonal components of the MAGNITUDE voltage.

### 3.4 Taking Input with the Arduino

**I**N this experiment, the Arduino microcontroller acts as an interface between the circuit and the computer. The Arduino's analog input pins are capable of sensing voltages between 0V and 5V, with a resolution just over 1mV. There are, however, some nuances related to both software and hardware if accurate readings are to be taken.

The analog input pins of the Arduino (A0 - A6) have very high open-circuit impedances. This means that even very small currents can cause significant voltage readings, particularly if no reference is given to the ground pin (GND). If the pins are left "floating", *i.e.* without a ground reference, then the input pins can measure the electromotive force induced by background radiation. For example, it was found that when a desk-lamp was turned on near the Arduino, appreciable voltages ( $\sim 2.5V$ ) were registered; a faulty ground connection was diagnosed as the cause of this problem.

Problems with the grounding of the Arduino are not limited to the GND pin itself. Faulty connectors from the circuit to the Arduino were also found to be an issue. In one case, connecting the Arduino to the lock-in amplifier via BNC cable caused the entire system to become grounded. A wire was then simultaneously connected from the output of the lock-in amplifier to an oscilloscope, which also showed that the voltage dropped to zero when the Arduino

was connected. This issue was resolved when the BNC cable between the lock-in amplifier and Arduino was replaced. It is likely that there was a short between the two wires within the cable, causing grounding of both cables when only one was connected to the GND pin.

In order to log the data measured by the Arduino, Excel's Data Streamer add-in was used. While other solutions do exist, Data Streamer was found to be most convenient both due to its simple user interface and the ease at which data can be quickly visualised in Excel — permitting the immediate rectification of any obvious problems. As the add-in outputs data in the csv format, a more rigorous data analysis can then be conducted at a later time through Python once no immediate problems are noted.

### 3.5 Taking Initial Measurements

As with the calibration of the interferometer (subsection 3.2), measurements were first taken with a laser as the light source, rather than an LED. Lasers are very highly monochromatic, and therefore have long coherence lengths (see Equation 9 and Figure 3c). This means that interference will appear even if the DMD between the paths in the Michelson interferometer isn't exactly zero. This makes them good candidates for testing the detection apparatus, as the existence of their interference pattern is only affected by the relative angle of the mirrors, which is easier to calibrate than the distance between the mirrors.

As a preliminary test, the micrometer was turned by hand with the angular velocity of the spindle being kept as constant as possible throughout. The micrometer was turned through a reading of 8.84 mm, which corresponds to an DMD range of just over  $9.4 \mu\text{m}$  due to the gearing ratio of the movable mirror's lever ( $1 \text{ mm} \rightarrow 1110 \text{ nm}$ ). The data collected from this measurement is displayed in Figure 5. The discrete Fourier transform (DFT) of the data is also displayed in Figure 7, with the peaks annotated.

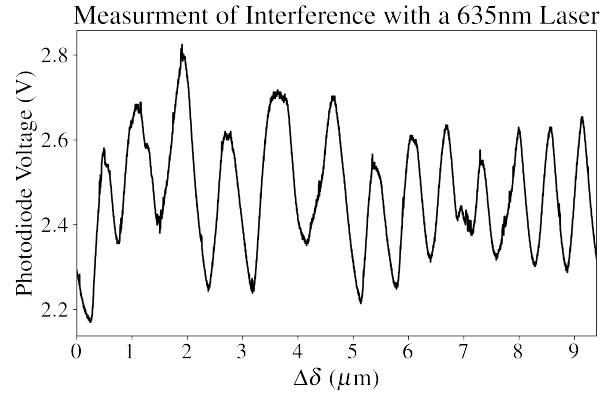


Figure 5: Coherence function of a red laser

The data was originally measured as a time series, and the  $x$ -axis was converted to a change in DMD ( $\Delta\delta$ ) under the assumption that the micrometer was moved at its mean angular velocity at all times. This is evidently not the case by looking at Figure 5; this is the reason for the second peak at 854 nm — it's erroneous. This can be shown definitively by splitting the domain of interference function and getting the DFT of both. In Figure 8, there are clearly two dominant frequencies in each partition. So, this is not an indicator of anything physical, rather it's an artefact of the inconsistent turning of the spindle.

The laser was then used to confirm the working order of the spectrometer. The SPD measured is displayed in Figure 6, a Gaussian profile has also been fitted. Through this regression, the wavelength of the laser is found to be  $(637 \pm 2) \text{ nm}$ . The wavelength quoted on the laser itself was 635 nm, and so the spectrometer is likely trustworthy enough to give reliable data about the LED.

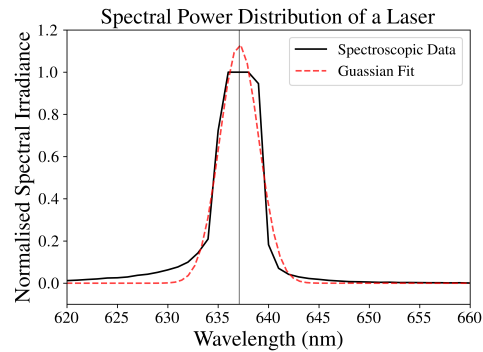


Figure 6: The SPD of the laser

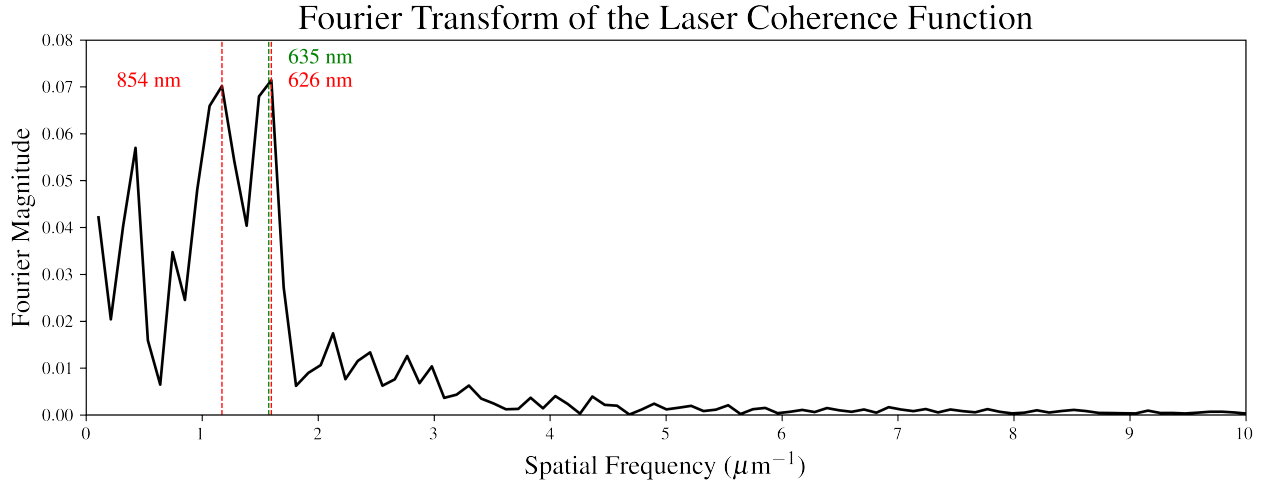


Figure 7: The Fourier transform of the laser's interference pattern reveals two dominant wavelengths, one of which is very close to the expected value.

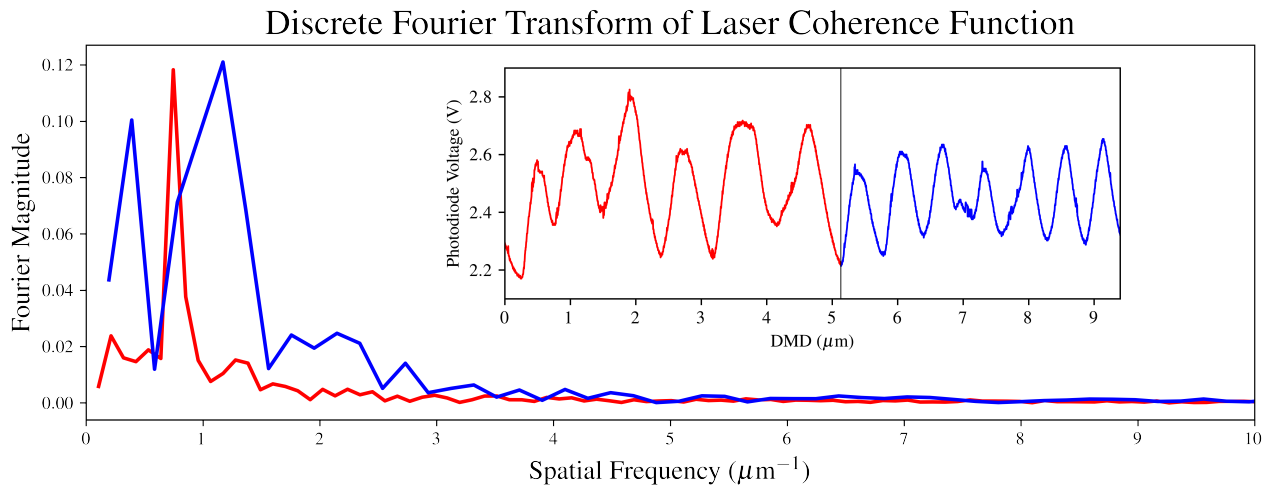


Figure 8: Splitting the coherence function into two partitions reveals two dominant frequencies at which the spindle was moved.

## 4 Results

### 4.1 Spectroscopic Analysis

THE spectrum of the LED was measured using the spectrometer immediately after that of the laser diode. The spectral power distribution measured is displayed in Figure 9, along with the corresponding Gaussian fit.

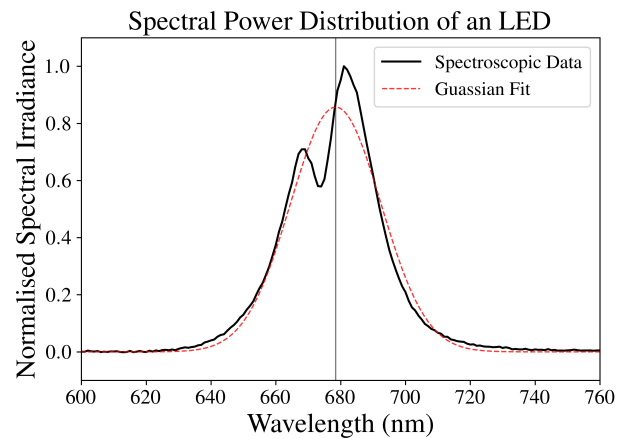


Figure 9: The SPD of the light emitting diode

Analysis of the spectral power distribution shows that the centroid wavelength  $\lambda_0$  is  $(678 \pm 14)$  nm, with a FWHM  $\Delta\lambda$  of  $(32.8 \pm 0.1)$  nm. Most LEDs produce an approximately normal spectrum (Kashiwao et al., 2025), and so the seemingly bimodal SPD of the LED used in this study is anomalous. A likely hypothesis is that the transparent material surrounding the LED is slightly absorbent to a narrow band of wavelengths around 670 nm.

It can be assumed that the LED would ideally have a Gaussian emission profile, and only deviates slightly in this experiment — justifying the normal fit in Figure 9. An equation for the coherence length, can be derived from Equation 11, as it is the DMD which causes the coherence envelope to decay to half of it's original magnitude (see subsection 2.1). This derivation is given in Appendix A.

$$L_c = \frac{2 \ln 2}{\pi} \cdot \frac{\lambda_0^2}{\Delta\lambda} \quad (12)$$

By using the data gathered from Figure 9 and Equation 12, a coherence length of  $6.2 \mu\text{m}$  was determined. Also present in Appendix A is the derivation for the uncertainty on the coherence length, Equation 13 is the result of this derivation.

$$\sigma_{L_c} = \frac{4 \ln 2}{\pi} \cdot \frac{\lambda_0}{\Delta\lambda} \sqrt{\sigma_{\lambda_0}^2 + \left(\frac{\lambda_0}{\Delta\lambda}\right)^2 \sigma_{\Delta\lambda}^2} \quad (13)$$

Upon substituting in the values obtained from the analysis of Figure 9 ( $\sigma_{\lambda_0} = 14$  nm,  $\sigma_{\Delta\lambda} = 0.29$  nm) into the above

equation, the uncertainty on the coherence length  $\sigma_{L_c}$  was found to be  $0.26 \mu\text{m}$ .

## 4.2 Interferometric Analysis

THE interference pattern obtained from the LED is displayed in Figure 10

Unlike the measurement of the laser (Figure 5), there is no strong oscillatory signal evident in the data — the changes in detected voltage having decreased by an order of magnitude too. The expected result, which is the coherence function of the LED with noise similar to that present in Figure 10, is plotted in Figure 11 by using the parameters found during the spectroscopic analysis (subsection 4.1).

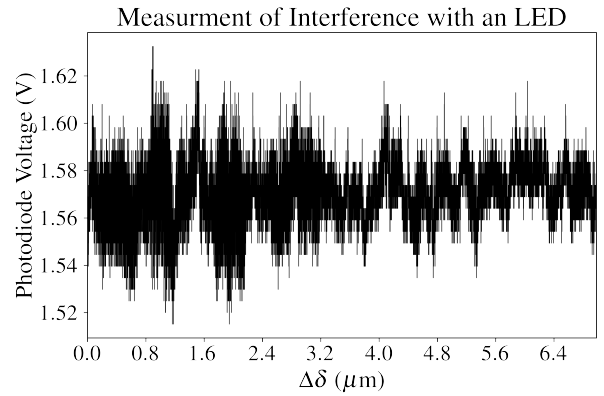


Figure 10: Interference pattern from the LED

The experimental data (Figure 10) does not show a strong similarity to the expected signal (Figure 11). This is shown quantitatively in Figure 12, where the expected and measured Fourier Transforms are compared.

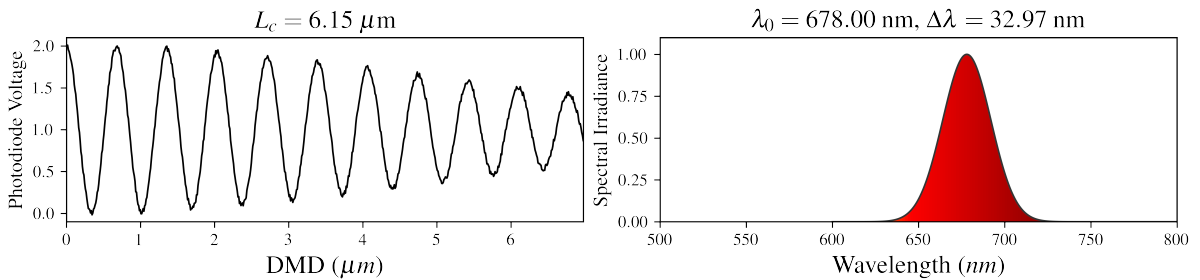


Figure 11: The expected coherence function of the red LED, with added noise of the same magnitude as in Figure 10. On the right is the Gaussian fit of the SPD displayed in Figure 9.

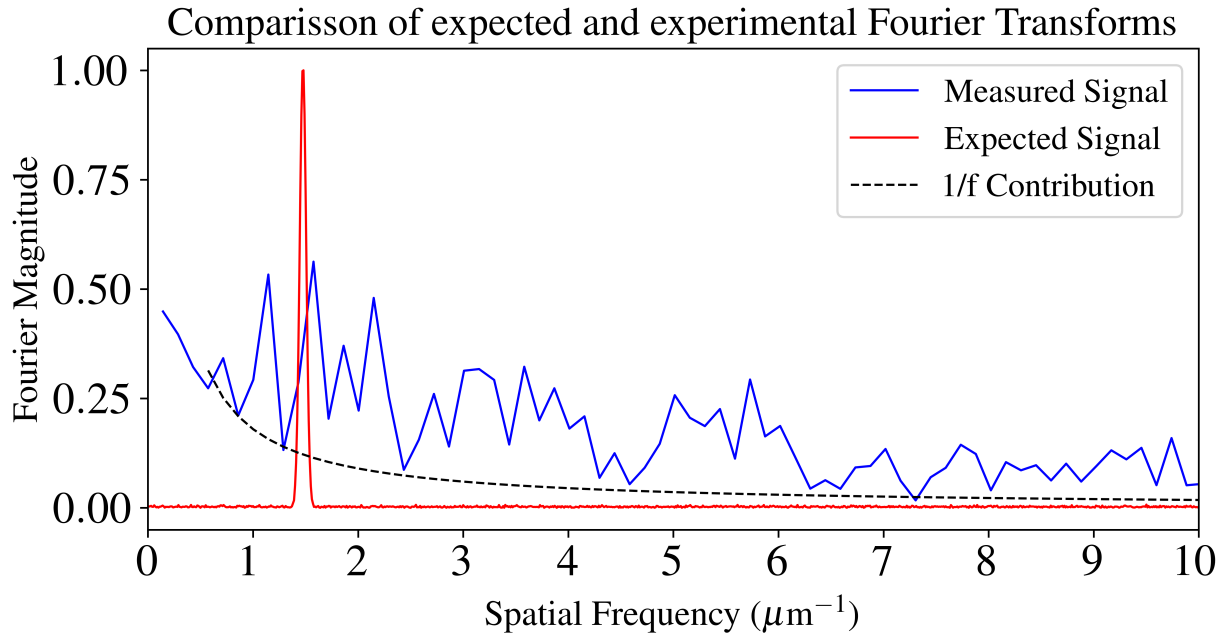


Figure 12: Comparison of the expected and measured signals. Also plotted is the contribution from 1/f noise (pink noise).

## 5 Discussion

### 5.1 The Deviation from Expectation

THE interference pattern measured from the light emitting diode was not similar to the theorised coherence envelope (see section 4). There is likely quite a simple explanation as for why this happened; the problem appears during the calibration of the interferometer.

While the multimeter has a gauge, this gauge can only measure relative distances — a zero-reading doesn't mean that each path in the interferometer is of equal length; rather a zero-reading acts as the origin of every other measurement. On the other hand, the coherence function depends on the *actual* difference in mirror displacement between the two paths which light travels along.

The calibration process for the interferometer is designed for lasers (Henschke, 2005), which have coherence lengths so long that the absolute path difference has almost no effect on the interference pattern. For a light source with a small coherence length, however, the mirrors must not only be

collimated but they also must be placed such that the two path lengths are within a coherence length of each other (in this case on the order of about  $10\ \mu\text{m}$ ). If the difference in mirror displacement is greater than this distance, then the photodiode will only operate in the constant luminosity region, which can be seen at the edges of Figure 3a and 3b. This would explain the lack of interference fringes present in Figure 10.

Placing the mirrors to within such a fine tolerance is a difficult task on its own, but that difficulty is made Sisyphean by the fact that the stationary mirror can't be displaced without also changing its angle, as it is moved via a pair of thumbscrews (Figure 4). This means that every time the mirror is translated, the collimation process must be repeated, but this again displaces the mirror slightly, and so on. Ultimately, attempting this calibration would be unreasonable without a method of translating the stationary mirror without changing its angle.

To verify that this hypothesis was indeed the reason for the failure of the interferometric mea-



surement, a screen was placed in front of the photodiode to catch the interference fringes with the laser as the source. As the micrometer was moved and the optical path difference increased, the interference fringes on the screen appeared to dilate. In fact, the micrometer was moved through its entire range and the fringes only ever expanded. If, however, the mirrors passed through the point of zero optical path difference, the interference fringes would have stopped dilating and started contracting instead — such a reversal was never observed, proving that the point of zero optical path difference was not within the range of the movable mirror. Without recalibrating the interferometer, interference fringes should not have been expected to appear for the LED. It would have been expected that these fringes should appear for the laser, however, as the coherence length of the laser is much longer than that of the light emitting diode. Both of these effects were observed in the experiment (Figure 10 and Figure 5 respectively).

## 5.2 Viable Solutions and Alternatives

The best way to acquire a path length difference of zero between the two arms of the interferometer avoids the problems with moving the fixed mirror mentioned in subsection 5.1. The end goal of creating no path length difference is to cause an equality in the phase of the two waves when they recombine at the beam splitter, and there exist methods which change the phase relation of the returning waves without touching the fixed mirror.

One such method, which is perhaps easiest to implement, involves changing the refractive index of the medium on one of the paths. It is known that a change in refractive index will cause a change in the wavelength of light, this in turn varies in the rate at which the phase of the wave changes per unit distance (*i.e.* the wavenumber). This means that the phase of one of the returning waves can be tuned by slowly varying the refractive index in one of the arms. If a screen is

placed to catch the interference fringes, then the variable refractive index can be swept through until those fringes reverse from contraction to expansion or vice versa (see subsection 5.1). (Albanelli et al., 2014) used a pressure cell to change the refractive index of air for a section of one of the interferometer paths. Furthermore, this source only used pressures below that of atmospheric pressure; if pressure was instead increased, then the phase relation could, in theory, be changed to any amount as long as the cell could contain large enough pressures.

Another method, suggested by (Conte et al., 2018), reflects one of the beams along a separate path, where a series of angled mirrors can be used to tune the optical path difference. This method is very similar to the experiment described in section 3.1, but with the advantage that the collimation process after selecting a set of positions for the mirrors doesn't change the selected optical path difference. This solution also forgoes the need to re-collimate the interferometer itself, rather only the series of extra mirrors must be collimated after each choice of optical path difference.

## 6 Conclusion

The value obtained for the coherence length of a red light emitting diode,  $(6.2 \pm 0.3) \mu\text{m}$ , agrees closely with other contemporary measurements. (Dubey et al., 2016) quotes a range of  $5.73 \mu\text{m}$  to  $6.92 \mu\text{m}$  for red and deep red LEDs, the result in this study falls within this range comfortably. (Mehta et al., 2010) performed measurements with a red LED which had a bandwidth of about half of that used in this study,  $\Delta\lambda = (17 \pm 1) \text{ nm}$ , and of slightly shorter wavelength,  $\lambda_0 = (641 \pm 1) \text{ nm}$ . It was found that localising the spectral power distribution and decreasing the centroid wavelength increased the coherence length of the LED to  $(24.18 \pm 4) \mu\text{m}$ , a four-fold increase from the value found in this study. This difference shows the sensitivity of the coherence length to the optical parameters of the LED.

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## Appendix A: Derivations<sup>1</sup>

### The Coherence Function — Equation 10

The on-axis intensity at the photodiode due to the interference of monochromatic light with wavenumber  $k$  and optical path difference  $\delta$  is given by (Humayun and Anwar, n.d.)

$$I(\delta, k) = 2B_k(1 + \cos k\delta) \quad (14)$$

Via Euler's formula  $e^{ix} = \cos(x) + i\sin(x)$ , Equation 14 can be written as

$$I(\delta, k) = 2B_k \left( 1 + \frac{e^{ik\delta} + e^{-ik\delta}}{2} \right) \quad (15)$$

$$\implies I(\delta) = 2B_k + B_k(e^{ik\delta} + e^{-ik\delta}) \quad (16)$$

In the above equations,  $B_k$  is the magnitude of the wavenumber  $k$  in the spectral power distribution of the light source. To show more explicitly that this is a function, it shall be written as  $B(k)$  from this point on. For polychromatic light, the total intensity at the photodiode is simply the integral of the intensity across all wavenumbers (due to the linear property of the Michelson interferometer discussed in subsection 2.5).

$$I_{\text{poly}}(\delta) = \int_{-\infty}^{\infty} I(\delta, k) dk \quad (17)$$

$$I_{\text{poly}}(\delta) = \int_{-\infty}^{\infty} 2B(k) + B(k)(e^{ik\delta} + e^{-ik\delta}) dk \quad (18)$$

$$\implies I_{\text{poly}}(\delta) = 2 \int_{-\infty}^{\infty} B(k) dk + \int_{-\infty}^{\infty} B(k)(e^{ik\delta} + e^{-ik\delta}) dk \quad (19)$$

Assume the LED has a Gaussian emission profile.

$$\therefore B(k) = B_0 \exp\left(-\frac{(k - k_0)^2}{2\sigma_k^2}\right) \quad (20)$$

Applying Equation 20 to Equation 19 produces

$$I_{\text{poly}}(\delta) = 2 \int_{-\infty}^{\infty} B_0 \exp\left(-\frac{(k - k_0)^2}{2\sigma_k^2}\right) dk + \int_{-\infty}^{\infty} B_0 \exp\left(-\frac{(k - k_0)^2}{2\sigma_k^2}\right) (e^{ik\delta} + e^{-ik\delta}) dk \quad (21)$$

The first integral is the improper integral of a Gaussian, which has the famous solution

$$\int_{-\infty}^{\infty} B(k) dk = \sqrt{2\pi} B_0 \sigma_k \quad (22)$$

The second integral can be split into two further integrals.

$$I_{\text{poly}}(\delta) = 2B_0\sigma_k\sqrt{2\pi} + \int_{-\infty}^{\infty} B_0 \exp\left(-\frac{(k - k_0)^2}{2\sigma_k^2}\right) e^{ik\delta} dk + \int_{-\infty}^{\infty} B_0 \exp\left(-\frac{(k - k_0)^2}{2\sigma_k^2}\right) e^{-ik\delta} dk \quad (23)$$

The remaining integrals are the Fourier and inverse Fourier transforms of the spectral power distribution  $B(k)$ . Even though the photons cannot interact with each other in time (subsection 2.5),

<sup>1</sup>All derivations are original, as no other similar derivations could be found.

these Fourier transforms provide the impression that the different wavenumbers of light *can* interfere with each other in  $\delta$ -space. Below is the solution to each integral,  $F_+$  and  $F_-$ .

$$F_{\pm} = \int_{-\infty}^{\infty} B_0 \exp\left(-\frac{(k-k_0)^2}{2\sigma_k^2}\right) e^{\pm ik\delta} dk \quad (24)$$

$$\Rightarrow F_{\pm} = B_0 \int_{-\infty}^{\infty} \exp\left(-\frac{k^2 - 2kk_0 + k_0^2}{2\sigma_k^2} \pm ik\delta\right) dk \quad (25)$$

$$\Rightarrow F_{\pm} = B_0 \exp\left(\frac{-k_0^2}{2\sigma_k^2}\right) \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2\sigma_k^2}k^2 + \left(\frac{k_0}{\sigma_k^2} \pm i\delta\right)k\right] dk \quad (26)$$

Use  $\int_{-\infty}^{\infty} \exp(-ax^2 + bx) dx = \sqrt{\frac{\pi}{a}} \exp\left(\frac{b^2}{4a}\right)$  with  $a = 1/2\sigma_k^2$  and  $b = k_0/\sigma_k^2 + i\delta$ .

$$\Rightarrow F_{\pm} = B_0 \exp\left(\frac{-k_0^2}{2\sigma_k^2}\right) \sqrt{2\pi\sigma_k^2} \exp\left[\frac{\left(\frac{k_0}{\sigma_k^2} \pm i\delta\right)^2}{4\left(\frac{1}{2\sigma_k^2}\right)}\right] \quad (27)$$

$$\Rightarrow F_{\pm} = B_0 \exp\left(\frac{-k_0^2}{2\sigma_k^2}\right) \sqrt{2\pi}\sigma_k \exp\left(\frac{k_0^2 \pm 2ik_0\sigma_k^2\delta - \sigma_k^4\delta^2}{2\sigma_k^2}\right) \quad (28)$$

$$\Rightarrow F_{\pm} = B_0 \exp\left(\frac{-k_0^2}{2\sigma_k^2}\right) \sqrt{2\pi}\sigma_k \exp\left(\frac{k_0^2}{2\sigma_k^2}\right) \exp(\pm ik_0\delta) \exp\left(\frac{-\sigma_k^2\delta^2}{2}\right) \quad (29)$$

$$\Rightarrow F_{\pm} = B_0 \sqrt{2\pi}\sigma_k \exp(\pm ik_0\delta) \exp\left(\frac{-\sigma_k^2\delta^2}{2}\right) \quad (30)$$

Inserting Equation 30 back into Equation 23 begets

$$I_{\text{poly}}(\delta) = 2B_0\sigma_k\sqrt{2\pi} + B_0\sqrt{2\pi}\sigma_k \exp(ik_0\delta) \exp\left(\frac{-\sigma_k^2\delta^2}{2}\right) + B_0\sqrt{2\pi}\sigma_k \exp(-ik_0\delta) \exp\left(\frac{-\sigma_k^2\delta^2}{2}\right) \quad (31)$$

$$\Rightarrow I_{\text{poly}}(\delta) = 2B_0\sigma_k\sqrt{2\pi} + 2B_0\sqrt{2\pi}\sigma_k \left(\frac{e^{ik_0\delta} + e^{-ik_0\delta}}{2}\right) \exp\left(\frac{-\sigma_k^2\delta^2}{2}\right) \quad (32)$$

Again, due to Euler's formula,  $\frac{e^{ik_0\delta} + e^{-ik_0\delta}}{2} = \cos(k_0\delta)$

$$\Rightarrow I_{\text{poly}}(\delta) = 2B_0\sigma_k\sqrt{2\pi} \left[1 + \cos(k_0\delta) \exp\left(-\frac{\sigma_k^2}{2}\delta^2\right)\right] \quad (33)$$

The sum of intensity contributed by each wavenumber must be the total intensity  $I_0$ . The spectral irradiance  $B(k)$  is that of the light coming from each arm of the interferometer, and is therefore contributes half the intensity of the light source itself. Mathematically, this can be expressed as

$$\frac{I_0}{2} = \int_0^{\infty} B(k) dk \quad (34)$$

The solution to this integral is present in Equation 22, as  $\int_{-\infty}^{\infty} B(k) dk \approx \int_0^{\infty} B(k) dk$  for  $k_0 \gg 0$  and  $\sigma_k \ll k_0$  (this is a *very* good approximation for an LED and other visible sources).

$$I_0 = 2\sqrt{2\pi}B_0\sigma_k \quad (35)$$

So, the coefficient out in front of the expression in Equation 33 is just the intensity at  $\delta = 0$ ,  $I_0$ . So, the coherence function  $I_{\text{poly}}(\delta)$  is given by

$$\Rightarrow I_{\text{poly}}(\delta) = I_0 \left[1 + \cos(k_0\delta) \exp\left(-\frac{\sigma_k^2}{2}\delta^2\right)\right] \blacksquare \quad (36)$$

### The Coherence Envelope — Equation 11

Expanding out the expression for the coherence function in Equation 36 gives

$$I_{\text{poly}}(\delta) = I_0 + I_0 \cos(k_0 \delta) \exp\left(-\frac{\sigma_k^2}{2} \delta^2\right) \quad (37)$$

This expression consists of a constant (which determines the intensity at  $\delta \gg L_c$ ), a cosine term (which introduces and controls the interference fringes) and a Gaussian term (which determines the luminosity of a given interference fringe).

It is this Gaussian term which suppresses the oscillations of the cosine term, and thus this term acts as the envelope on the coherence function — it is the coherence envelope.

$$I_{\text{env}}(\delta) = I_0 \exp\left(-\frac{\sigma_k^2}{2} \delta^2\right) \quad \blacksquare \quad (38)$$

### The Coherence Length— Equation 12

The coherence length is defined as the DMD at which the coherence envelope is half its maximum.

$$\frac{I_0}{2} = I_0 \exp\left(-\frac{\sigma_k^2}{2} L_c^2\right) \quad (39)$$

$$\implies \ln\left(\frac{1}{2}\right) = -\frac{\sigma_k^2}{2} L_c^2 \quad (40)$$

$$\implies -\ln 2 = -\frac{\sigma_k^2}{2} L_c^2 \quad (41)$$

$$\implies L_c^2 = \frac{2 \ln 2}{\sigma_k^2} \quad (42)$$

The full-width at half-maximum (FWHM) in terms of wavenumber  $\Delta k$  is related to the standard deviation  $\sigma_k$  by

$$\Delta k = 2\sqrt{2 \ln 2} \sigma_k \quad (43)$$

Let  $k_1$  and  $k_2$  be the wavenumbers at the edge of the FWHM, *i.e.*  $\Delta k = |k_2 - k_1|$ .

$$\Delta k = \left| \frac{2\pi}{\lambda_2} - \frac{2\pi}{\lambda_1} \right| \quad (44)$$

$$\implies \Delta k = 2\pi \frac{|\lambda_1 - \lambda_2|}{\lambda_1 \lambda_2} \quad (45)$$

If  $\lambda_0 \gg \Delta \lambda$ , as is the case for the sources in this study, then  $\lambda_1 \lambda_2 \approx \lambda_0^2$

$$\implies \Delta k = 2\pi \frac{\Delta \lambda}{\lambda_0^2} \quad (46)$$

Then, using Equation 43

$$2\sqrt{2 \ln 2} \sigma_k = 2\pi \frac{\Delta \lambda}{\lambda_0^2} \quad (47)$$

$$\Rightarrow \sigma_k = \frac{\pi}{\sqrt{2 \ln 2}} \frac{\Delta \lambda}{\lambda_0^2} \quad (48)$$

Filling this conversion into Equation 42 gives

$$L_c^2 = 2 \ln 2 \left( \frac{\sqrt{2 \ln 2}}{\pi} \frac{\lambda_0^2}{\Delta \lambda} \right)^2 \quad (49)$$

$$\Rightarrow L_c^2 = \left( \frac{2 \ln 2}{\pi} \frac{\lambda_0^2}{\Delta \lambda} \right)^2 \quad (50)$$

$$\Rightarrow L_c = \pm \frac{2 \ln 2}{\pi} \frac{\lambda_0^2}{\Delta \lambda} \quad (51)$$

As the coherence length is a length, it's defined as being explicitly positive.

$$\therefore L_c = \frac{2 \ln 2}{\pi} \frac{\lambda_0^2}{\Delta \lambda} \quad \blacksquare \quad (52)$$

### The Coherence Length Uncertainty — Equation 13

If  $\{x_0, \dots, x_n\}$  is a set of measurable observables, and  $f(x_0, \dots, x_n)$  describes a quantity which can be calculated from those observables, then the uncertainty on  $f$  is given by taking the effect of each variable on the function (*i.e.* the partial derivative with respect to that variable) and adding the uncertainty due to that effect in quadrature.

$$\sigma_f = \sqrt{\sum_{i=0}^n \left( \frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2} \quad (53)$$

Applying Equation 53 to Equation 52 will give the uncertainty on the coherence length.

$$\sigma_{L_c} = \sqrt{\left( \frac{\partial L_c}{\partial \lambda_0} \right)^2 \sigma_{\lambda_0}^2 + \left( \frac{\partial L_c}{\partial (\Delta \lambda)} \right)^2 \sigma_{\Delta \lambda}^2} \quad (54)$$

$$\Rightarrow \sigma_{L_c} = \sqrt{\left( \frac{4 \ln 2}{\pi} \frac{\lambda_0}{\Delta \lambda} \right)^2 \sigma_{\lambda_0}^2 + \left( \frac{-4 \ln 2}{\pi} \frac{\lambda_0^2}{(\Delta \lambda)^2} \right)^2 \sigma_{\Delta \lambda}^2} \quad (55)$$

$$\Rightarrow \sigma_{L_c} = \frac{4 \ln 2}{\pi} \frac{\lambda_0}{\Delta \lambda} \sqrt{\sigma_{\lambda_0}^2 + \left( \frac{\lambda_0}{\Delta \lambda} \right)^2 \sigma_{\Delta \lambda}^2} \quad \blacksquare \quad (56)$$

## Appendix B: Reflection

*Note: Any first-person pronouns in this appendix are referring to the author, Adam Quayle.*

Over the course of the project, everyone contributed valuable effort according to their own strengths. Max and I worked primarily on the theory (section 2) and analysis (section 4). Patrycja and Juliet contributed strongly to the technical details of the experiment including the tuning of the lock-in amplifier (subsection 3.3) and the drafting of the risk assessment (Appendix C).

The experimental portion of the study encountered many issues, the most time-consuming being the tuning of the lock-in amplifier. While the manual was combed through many times, the group took almost two lab sessions (~5 hours) to tune the amplifier to an optimal state. In retrospect, many more diagnostic tests should have been preformed — rather than trying to follow the manual to the word. Doing these tests would have brought us naturally closer to the maximal gain state of the amplifier. The lack of progress regarding the lock-in amplifier ultimately subtracted from the time available to take actual measurements. One positive is that the team remained co-operative and driven during this arduous process.

From my point of view, the most regrettable aspect of this project is that the fundamental error in calibration mentioned in subsection 5.1 wasn't noted sooner, perhaps due to the lost time spent in calibrating the lock-in amplifier. In subsection 5.2, a possible solution regarding a pressure cell was discussed. Perhaps this alternative would have been discovered and pursued if time allowed, but with the only method of gathering data being reduced to spectroscopy (subsection 2.6) all effort was put into obtaining that data. This missed opportunity is made more painful by the fact that there was a vacuum cell sitting on the lab-bench throughout the four weeks this experiment was conducted. Furthermore, the only other group present in the lab was preforming an experiment which used a vacuum cell to cause the exact corrective change in phase which was discussed in subsection 5.2 — and were successful in doing so.

While the experiment did not run as smoothly as the group would have wished, the group was still happy with the experience and how we worked together. Max and I analysed the spectroscopic data, which went without issue and everyone in the group contributed to the group presentation. I compiled everybody's slides into the final presentation and was impressed by the effort put in by each party.

Unfortunately, it was only after the presentation that I was able to complete my derivation of the coherence function (Appendix A) after the integral I was stuck on happened to show up in a quantum mechanics question which I had solved. This meant that all of the graphics which involved coherence functions for the presentation had to be generated computationally, each taking about 10 hours to render due to the multiple integrals that had to be done per step. For comparison, the coherence functions plotted in this paper each took about 10 *seconds* to render.

After all that's been said, I am very happy with how the group worked together and wouldn't have changed anything about our dynamic. While problems did appear, the group was always willing to discuss and try to make progress towards our goal — which we did achieve, albeit in a method not originally intended.



## Appendix C: Risk Assessment

Below is the revised risk assessment of the experiment detailed in this paper, which was created at the outset of the project by Patrycja and Juliet, and was approved by the rest of the group.

**ASSESSMENT DATE:** 18/02/2026

**TITLE:** Measurement of the Coherence Length of a Light Emitting Diode

**LOCATION:** Optics Lab, Department of Physics, Maynooth

**DESCRIPTION OF PROJECT / EXPERIMENT:**

The determination of the coherence length of a light emitting diode by measuring the fringes of it's interference pattern via the Michelson interferometer. A lock-in amplifier is used to amplify the low-tension signal from the photodiode and this amplified signal is recorded via an Arduino Uno.

**UNIVERSAL CONTROLS:**

No food and drink in the lab. Bags and chairs tucked under desk. Small lamp stays illuminated except for when the interferometer is being used. Equipment is turned off when not in use.

**PPE REQUIRED:** N/A

### Lists of Hazards

**HAZARD TYPE:** Ocular damage while looking through the interferometer eyepiece.

**HAZARD DESCRIPTION:**

Laser being used to calibrate the interferometer is at a high intensity. Looking directly at the laser can cause damage to the eye. Looking at low-intensity light on the screen for an extended time may cause eye strain.

**CONSEQUENCE:** Damage to the eye or eye strain.

**WHO IS AT RISK:** The observer.

**CONTROL MEASURES ALREADY IN PLACE:**

Laser is projected onto a screen where it will be observed, eliminating the danger of looking directly into the laser. Laser is at waist level to avoid accidentally looking into it. The mirrors of the interferometer are collimated such that no stray light should exit into the surroundings.

**RISK RATING:** L.

**ACTION PLAN:** If eye is exposed to high-intensity light, avert and seek treatment.

**HAZARD TYPE:** Diode combustion

**HAZARD DESCRIPTION:**

This experiment consists of working with diodes. This means that there is a risk of fire or damage if the polarity is incorrect or voltage is too high.

**CONSEQUENCE:** Damaged diode.

**WHO IS AT RISK:** The diode.

**CONTROL MEASURES ALREADY IN PLACE:**

Wires connected to diode should be labelled with a maximum voltage.

**RISK RATING:**M.

**ACTION PLAN:**

Check voltage prior to switching on the circuit. Turn off circuit immediately if it starts emitting a burning smell or smoke. Check with the demonstrator or technician on next steps rather than fixing it yourself or turning it back on.